Next-Gen Solid-State Battery Core Technology Whitepaper

System-level solution based on diamond-reinforced carbon skeleton and co-sintered monolithic architecture

Industry Background & Technical Challenges

All-solid-state batteries (ASSBs) are seen as a key technology to break through the energy density ceiling of current lithium-ion batteries. However, in the transition from lab to scale, the industry faces dual bottlenecks at both the "material" and "structural" levels.

Dendrite Penetration & Short Circuit

During charging, needle-like dendrites easily form on the lithium metal anode surface. Existing carbon host materials

Interface Contact Failure

Electrode materials undergo large volume changes (breathing effect) during charge/discharge. Traditional physically stacked solid-solid interfaces

typically have modulus < 10 GPa, unable to physically block dendrite growth, leading to electrolyte penetration and short circuit.

are prone to delamination, causing interface impedance (R_{ct}) to rise sharply.

🛑 Complex Manufacturing

Cathode, electrolyte, and anode are typically prepared separately and then assembled; the process chain is long, and scaling ultrathin electrolyte layers is difficult.

Base Technology: High-Modulus Anode Host Material

Product A: Diamond/Carbon Composite Anode Host

Core Material

Addressing the pain points of "dendrite penetration" and "volume expansion" in lithium metal anodes, we developed a 3D skeleton material based on **sp^3-sp^2 covalent bonding network**. This forms the foundation of our entire technology system.

💎 High Modulus (>100 GPa)

Nano-diamond is introduced as rigid "rivets" to build a physical "dendrite exclusion zone," effectively mitigating short-circuit risks.

🔋 Zero-Strain Lithium Storage

Rich internal connected pores accommodate metal lithium deposition, achieving near-zero volume expansion during charge/discharge.

⚡ **Gradient Lithiophilic Design**

In-situ doping (B/N/Si) induces lithium ions to preferentially deposit inside the skeleton, achieving uniform dendrite-free lithium plating.

Advanced Technology: Integrated Gradient Skeleton Structure

PRODUCT B

Monolithic Integrated Skeleton

Building on Product A's material advantages, we further propose an innovative **"zoned functionalization + monolithic co-sintering"** architecture, fusing cathode, electrolyte, and anode matrix into a single rigid block.

- **Seamless interface:** Cathode, electrolyte, and anode skeletons form an atomically bonded continuous whole through co-sintering, effectively eliminating physical seams; interface impedance approaches liquid battery levels.
- **Pore gradient design (see right):**
  - Electrode zones: **micron-scale macropores** for high-capacity active material loading.
  - Electrolyte zone: **nanoscale mesopores** for electrolyte self-adsorption and locking via capillary pressure difference; physically blocks cathode-anode interpenetration.
- **Triple insulation strategy:** Insulation coating or graded composition achieves electronic insulation while maintaining mechanical continuity.

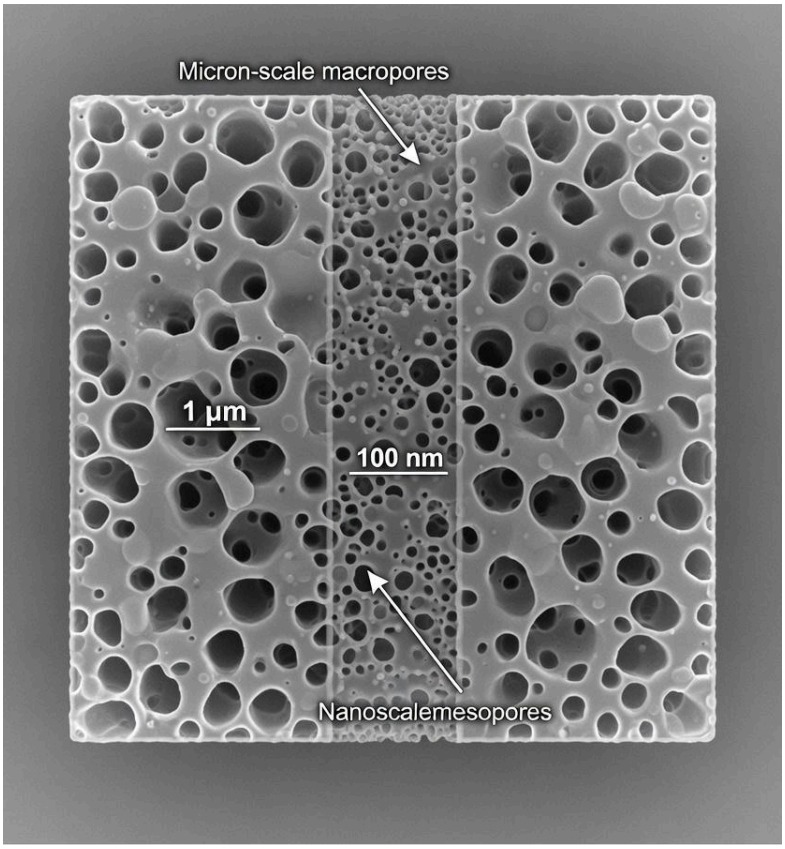

Fig 1: Monolithic co-sintered skeleton microstructure (SEM)

Micron pores on both sides • Nano mesopores in center • No physical interface

Potential Application Advantages & Value

📱 Consumer Electronics (3C)

Eliminate "swelling" anxiety: Zero-strain skeleton enables device makers to avoid reserving physical space for battery expansion, significantly increasing capacity while keeping thickness unchanged.

Better fast-charging: sp^2 conductive skeleton provides excellent electron transport channels, supporting high-rate fast charging without overheating.

🚁 Low-Altitude Economy (UAV/eVTOL)

High energy density: Enables lithium metal anode, significantly increasing gravimetric energy density (Wh/kg), directly translating to longer flight time and range.

Structural safety: The monolithic rigid skeleton itself serves as a structural component; maintains cell integrity under drop or impact, mitigating fire/explosion risk.

🚗 New Energy Vehicles (EV)

Simplified pack process: High mechanical strength of single cells allows simplified external pressure structure for Battery Pack, improving GCTP integration efficiency.
Manufacturing cost reduction: Supports "ingot sintering + wire cutting" process similar to PV industry; potential to significantly reduce solid-state battery manufacturing cost.

🔋 Grid-Scale Storage

Long cycle life: Rigid skeleton limits mechanical pulverization of electrode materials during repeated charge/discharge; theoretical cycle life far exceeds traditional lithium-ion batteries.
Inherent safety: No flammable liquid components; effectively addresses thermal runaway risk in energy storage stations.

Nanjing CuFeng Mechanical & Electrical Technology Co., Ltd.

Address: Room 909, Building 2, No. 2 Xingzhi Road, Jiangbei New Area, Nanjing, Jiangsu | Email: wangbo@tospike.com | Web: www.tospike.com

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